

THE INCIDENCE AND DURATION OF COLLUSION: THE CASE OF US GENERIC PRESCRIPTION DRUGS

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MOTIVATION

- There has been increasing concern about lack of competition
- Policy tools include regulation and antitrust
- Primary components of antitrust policy
 - Merger review
 - Abuse of dominance cases
 - Prosecution of collusive agreements
- Cartels fundamentally threaten competition
- Generic prescription pharmaceuticals, a prominent US price fixing case, provide an informative case study

COLLUSION

- Recent US DOJ price fixing cases
 - All 3 major canned tuna brands
 - 70 auto parts suppliers
 - 15 global banks trading foreign exchange
 - "Big pork" and "big chicken"
 - 100+ real estate investors involved in foreclosure auctions
 - Many others—including the subject of today's talk
- *Which markets are cartelized?*
- Entry provides natural safeguard [Stigler, 1964]
- Yet, entry is often slow and/or expensive
- *What is the extent and effect of entry in cartelized markets?*

REFERENCES

- This talk is based on:
 - E. Cuddy, R. Porter, A. Starc, and T. Wollmann, Entry Barriers, Personal Relationships, and Cartel Formation: Generic Drugs in the United States, in J. Harrington and M.P. Schinkel, eds., *Cartels Diagnosed: New Insights on Collusion*, Cambridge, 2024
- which in turn relies on:
 - Emily Cuddy, Competition and Collusion in the Generic Drug Market
 - Amanda Starc and Tom Wollmann, Does Entry Remedy Collusion? Evidence from the Generic Prescription Drug Cartel
 - *Connecticut et al. v. Teva Pharmaceuticals USA et al.* (20 May 2019)
 - *I assume the facts reported in the Complaint are correct; formally, I describe "alleged" behavior w/r/t certain agents*

SUMMARY

- Study the US generic prescription drug cartel (possibly the largest price fixing case in US history)
- Exploit detailed information about internal operations
- Measure the effect of cartel formation on prices and entry
- Assess how the cartel's operations and market outcomes conform with received theories of collusion

COLLUSIVE AGREEMENTS

A collusive scheme must:

- Enlist conspirators
- Come to an agreement (on prices, market division, etc.)
- Enforce agreement (detect and respond to deviations)
- Protect gains from entry or expansion by non-participants
- Respond to new circumstances
- Evade detection by victims and antitrust agencies

How did the generic drug cartel operate?

AGENDA

- Background
- Data
- Evidence
- Analysis
- Lessons for policy

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Background

MARKET OVERVIEW

- Generic drugs account for >90% of prescription fills in US
- Branded drug loses exclusivity → ANDAs → generic entry
 - Generics bioequivalent to patented drug, require FDA approval
 - ANDA: Abbreviated New Drug Application
 - ANDAs apply to all strengths within a substance-delivery-release
- Typically heralded as competition success story
(>50% generic market share at >50% lower prices)
- Generic manufacturers sell drugs to pharmacy chains, wholesalers
 - Adoption of auctions in years before cartel formation
- Most consumers' insurance coverage includes prescription drug benefit
- Insurance companies reimburse pharmacies for each fill
- Prices of many drugs increased substantially in the 2010s

CARTEL FORMATION

- Inner workings described in Complaint (filed 5/2019, unsealed 6/2019)
- At least 19 generic producers joined conspiracy
- Formed bidding ring
 - Bidders differed across markets due to different drug portfolios
- Allocated market share across buyers to attain "fair share"
- 4/2013: Teva hires NP, tasks with "price increase implementation"
- Extraordinary nature not lost on industry leaders
- 1000s of conversations among rivals; cartels form in ~ 150 markets (i.e., explicit collusion)
- Commonly prescribed drug prices rise sharply [Cuddy 2020; Clark, Fabiilli, and Lasio 2022]

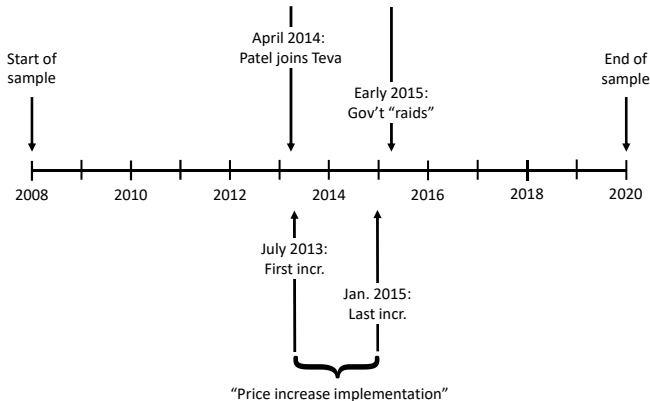
FACTORS CONDUCTIVE TO COLLUSION

- Firms compete mostly on price in procurement auctions
- Cartel members could detect cheating quickly
 - In 2013, Glenmark accidentally undercut Teva in one market
 - Later the same day, a confused colleague emailed NP "???"
 - 5 minutes later, NP replied, "I know...made the call already"
 - The next day, NP spoke to her counterpart at Glenmark
 - Later that day, Glenmark withdrew the lower price
- Defections could be punished quickly
 - Procurement auctions spread throughout year
- Demand is relatively inelastic and stable
- Multimarket contact
- Firms have similar costs
- Entry is slow and expensive

INVESTIGATION, LITIGATION, PROSECUTION

- Mid-2014: CT AG opens investigation
- Early-2015: government "showing up on people's doorsteps"
- Early-2017: first lawsuits alleging Section 1 violations
- Early-2025: cases in various stages of litigation/prosecution
- Impact on behavior
 - No additional markets cartelized after 2015 raids
 - Existing agreements probably in place today, albeit tacitly (Starc and Wollmann test using Backus, Conlon, and Sinkinson [2021] framework)
 - "Collusion is hard to initiate, easy to sustain"

TIMELINE



Data

SOURCES

- *Connecticut et. al. v. Teva Pharmaceuticals USA et. al.* (May 20, 2019)
 - List of cartelized drugs
 - Date that each drug market's cartel was formed
 - Background information
 - NP's selection rule
 - Phone records, emails, text messages
 - Teva's market share estimates drug-firm (to validate data)
- FDA: ANDA filings, approvals, and entry dates
- Medicaid and IQVIA: Price and quantity

DATA STRUCTURE

- Unit-of-observation
 - Drug-firm-quarter or drug-firm-year
 - "Drug" $\equiv$ substance-delivery-release-strength
 - "Firm" $\equiv$ ultimate parent entity
- Sample
 - Covers 2008 – 2020
 - Comprises drugs manufactured by Teva at Q1 2013
 - Excludes drugs going off-patent in/after 2008

BRIEF SUMMARY

Drugs in sample: 416

Cartelized drugs: 113

Proportion cartelized: 27%

Price per prescription: \$21.7 [\$35.8]

Expend. per drug per year: \$30.2MM [\$44.3MM]

Firms per market (YE 2012): 5.1 [4.1]

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Evidence

WHICH DRUGS WERE CARTELIZED?

- NP compiled list of Teva drugs (i.e., sub.-delivery-release-strength)
 - Scores competitors by "quality" on scale from -3 to 3; based on strength of personal relationship
 - Averages quality for each drug, weighted by market share
 - Also considers number of active competitors in market
 - Ranks drugs to select "Immediate Price Increase File"
- Cartels form in 109 drug markets
- Another 30 drug markets added over next 18 months
 - Personnel changes (e.g., KG moves to Zydus)
 - Exogenous exit vis-a-vis supply disruptions
 - Would have rolled out faster but for NP's maternity leave

DETERMINANTS OF COLLUSION

Competition Quality	Number of active firms			
	1 other	2 others	3 others	4+ others
{-3,-2,-1}	15% (13)	15% (20)	15% (20)	15% (27)
0	22% (9)	38% (21)	31% (13)	12% (33)
1	57% (7)	50% (24)	28% (18)	17% (23)
2	50% (8)	57% (14)	67% (6)	0% (6)
3	85% (13)	89% (9)	60% (5)	0% (1)

SPECIFICATION(S)

- Variation
 - Cartelized v. uncartelized drugs
 - Before v. after cartel formation
- Presentation
 - Raw data in calendar time
 - Event study estimates (for legibility)

Y is outcome; χ indicates cartel drug-time observation.

Report estimates from

$$Y_{dt} = \sum_{\tau=-21}^{26} \beta^{\tau} \chi_{dt}^{\tau} + \theta_d + \phi_t + \epsilon_{dt}. \quad (1)$$

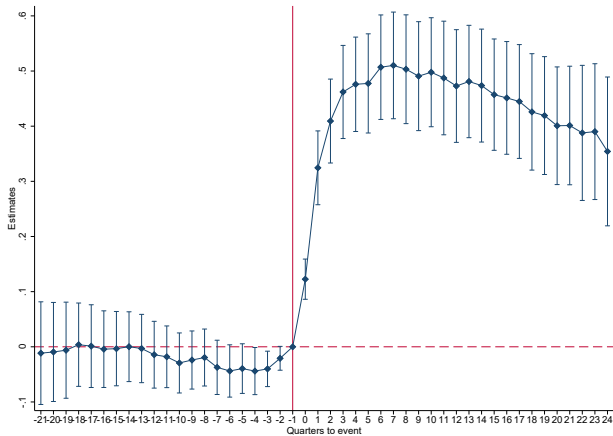
Sun and Abraham [2020] approach yields similar results.

How much do prices change?

PRICES IN CALENDAR TIME

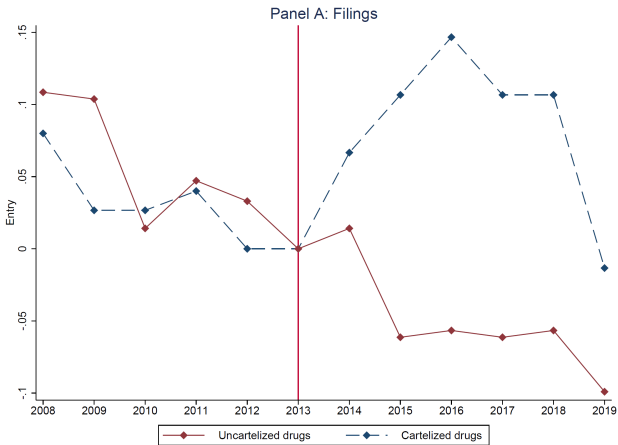


PRICES IN EVENT TIME

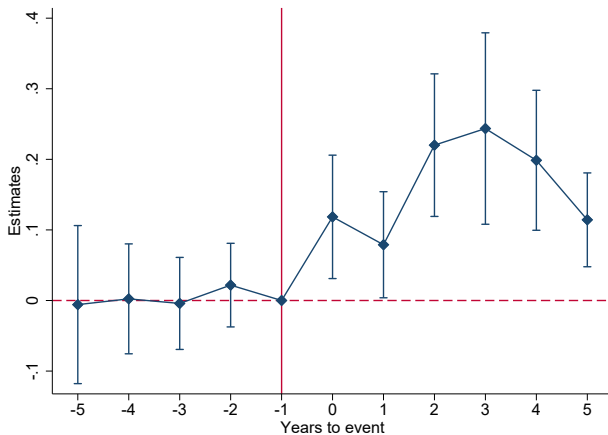


Does cartel formation attract entry?

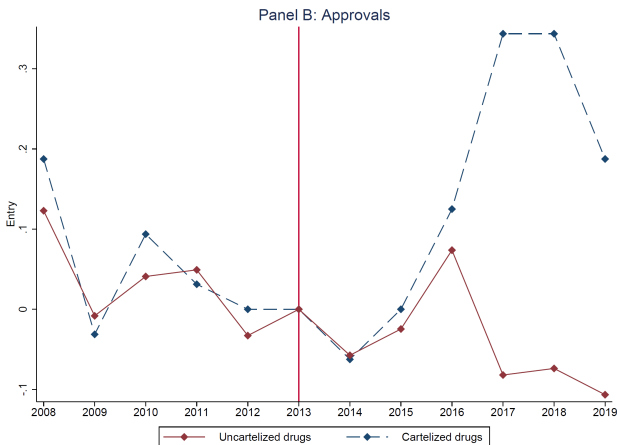
ENTRY (ANDA FILINGS) IN CALENDAR TIME



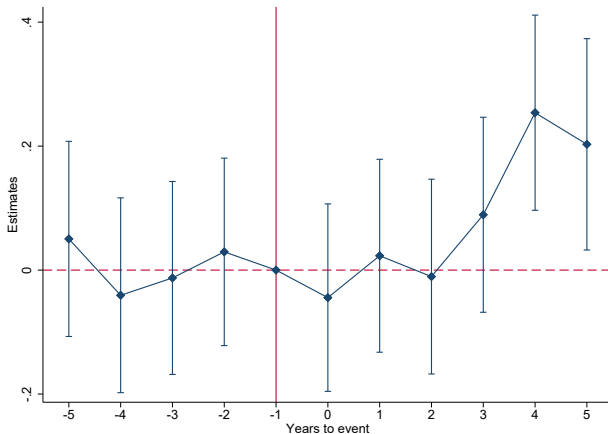
ENTRY (ANDA FILINGS) IN EVENT TIME



ENTRY (ANDA APPROVALS) IN CALENDAR TIME

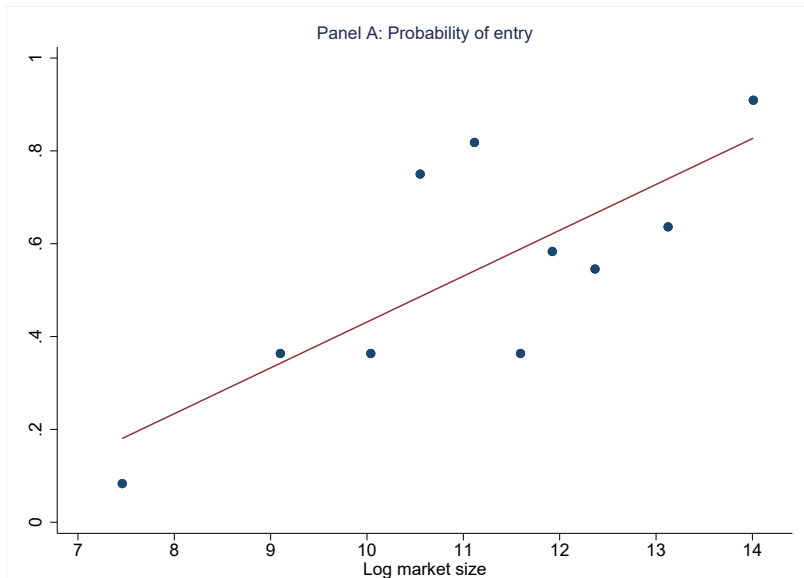


ENTRY (ANDA APPROVALS) IN EVENT TIME



Does entry reduce price?

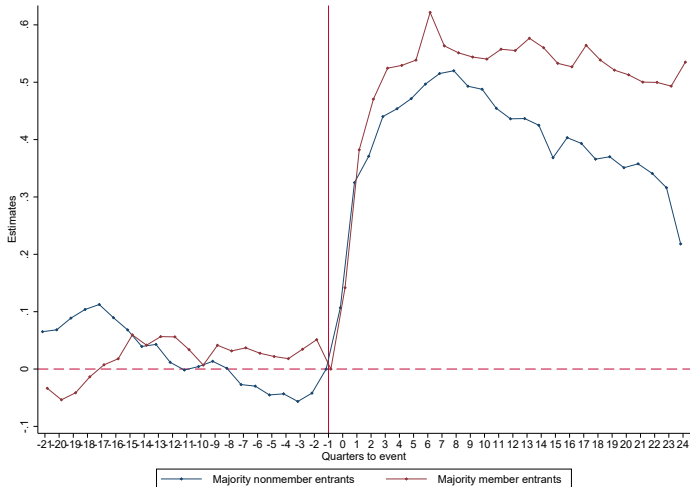
LARGER MARKETS ATTRACT MORE ENTRANTS



PRICES IN CALENDAR TIME BY MARKET SIZE

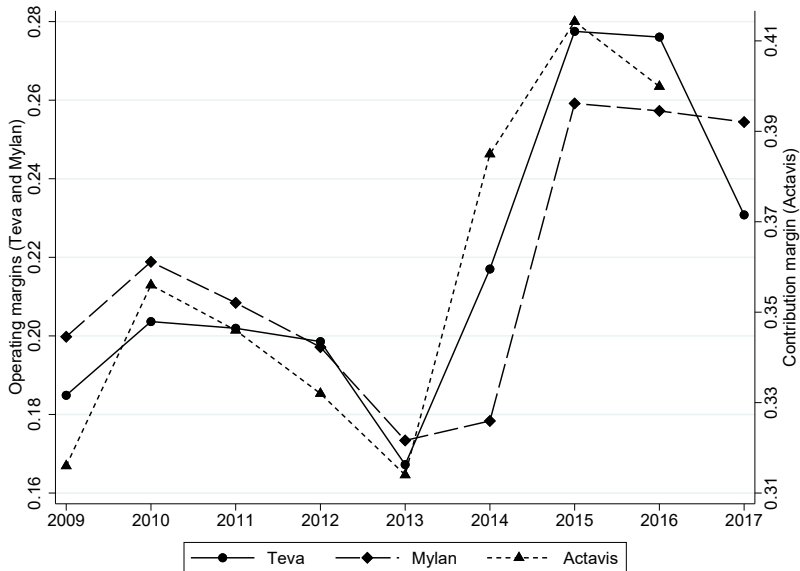


PRICE SERIES BY TYPE OF ENTRANT



What happened to profit margins?

MARGINS INCREASED, REMAINED HIGH



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Analysis

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AGREEMENT AND ENFORCEMENT

- Agreement
 - Cartel allocated buyers to achieve "fair" share
 - Unusual in that price increases were abrupt
 - Unlikely to be joint profit maximizing, even ignoring legal risk
- Enforcement
 - Agreement must be self-enforcing
 - All members must find cheating unprofitable
 - Detection and retaliation *rapid* in this setting
 - Multimarket contact strengthens punishment
 - Incentive compatibility constraints are *almost trivially* satisfied

EVASION

- Detection by victims
 - Consumers did not bear higher costs in short run
 - Insurance companies reimbursement of pharmacies based on lagged price indices
 - Indices slow to reflect higher prices
 - In short run, losses borne by direct purchasers
- Antitrust enforcement
 - Abrupt price hikes reported in NYT, spurred CT investigation
 - Increased likelihood of antitrust action
 - No further cartelization attempts once detected

PASS THROUGH EFFECTS

- Under reimbursement rules, costs borne by insurers in medium run
- Before 2014, generic prices fell, branded prices rose
- Nevertheless, some concern about increasing concentration in generic segment
- Insurers responded to price hikes by:
 - increasing consumer's out of pocket costs, deductibles, or premia
 - switching to coinsurance
 - changing the list of covered drugs
- Potential for consumer demand response to insurance changes
 - Demand side response may discipline cartel price increases
 - But potential harm from health consequences

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Conclusions

TAKEAWAY LESSONS AND FUTURE WORK

- Does entry discipline the cartels?
- Yes (entrants arrive and exert negative pressure on price) and no (entry is delayed and incomplete)
- Reducing regulatory costs/delays improves efficiency
- Section 1 enforcement serves a critical role
- Much more work needed in this area (e.g., in light of recent cases, whether industry-wide consolidation facilitates coordination, data-based screens, "breaking" ongoing cooperation, "relationships")

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Thank you